

Introduction to Amazon DynamoDB: Hands-on Lab

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Lab overview

In this lab, I learned how to create a table in Amazon DynamoDB to store details about a music library. I practiced running queries on the library and then deleted the table at the end.

Objectives

- Create an Amazon DynamoDB table.
- Enter data into an Amazon DynamoDB table.
- Query an Amazon DynamoDB table.
- Delete an Amazon DynamoDB table.

Task 1: Create a New Table

- Step 1: Search and Choose “DynamoDB” from the AWS Management Console

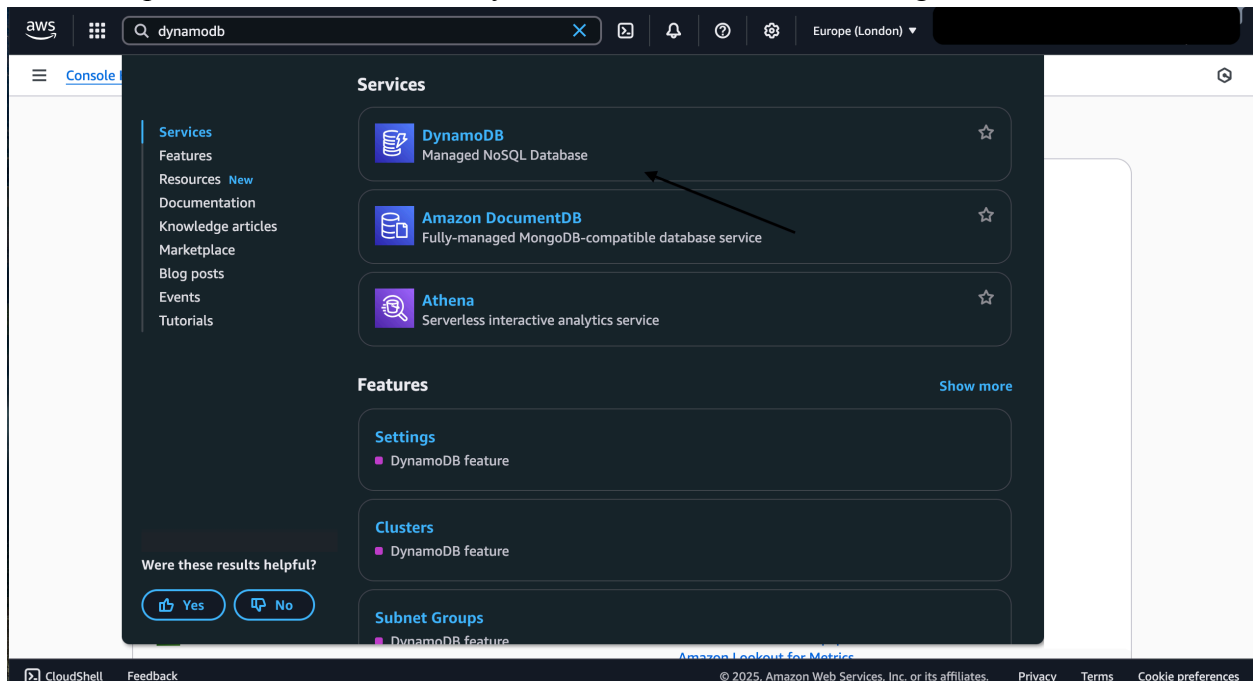


Figure 1: DynamoDB search (source: personal collection)

- Step 2: From the starting page, choose “Create table”

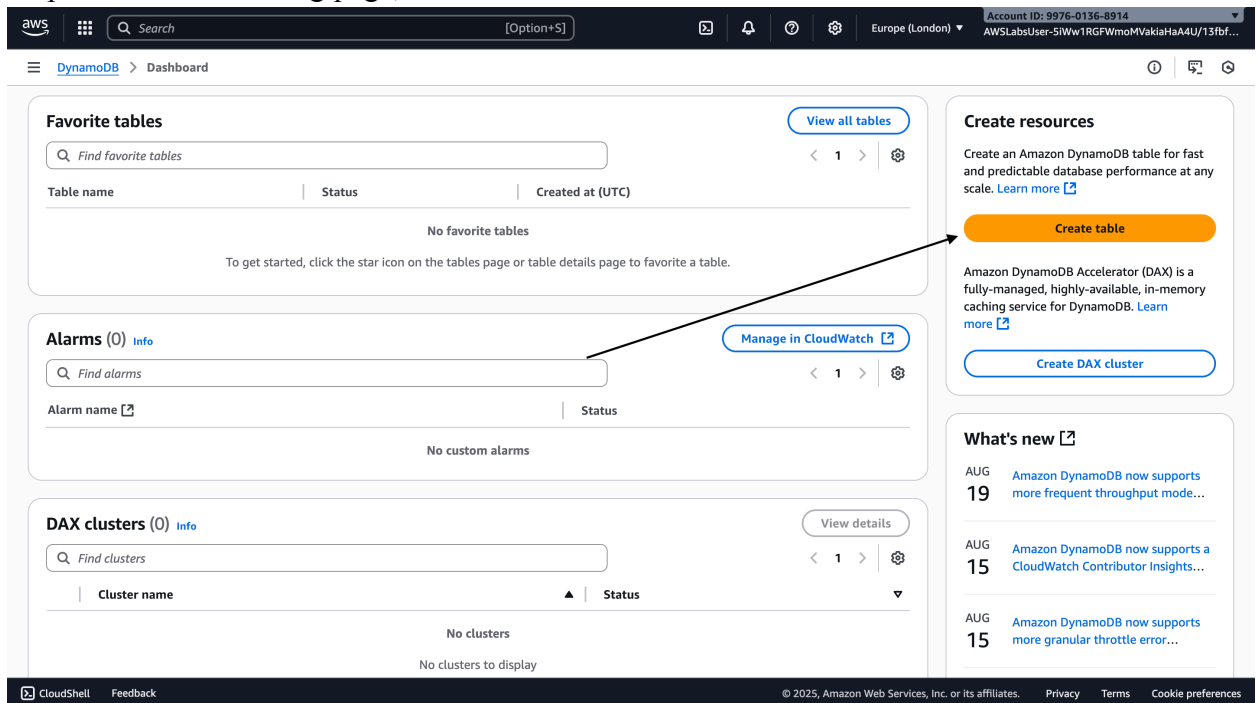


Figure 2: Create table (source: personal collection)

- Step 3: On the Create table page, configure the Table name, Partition key, and Sort key.

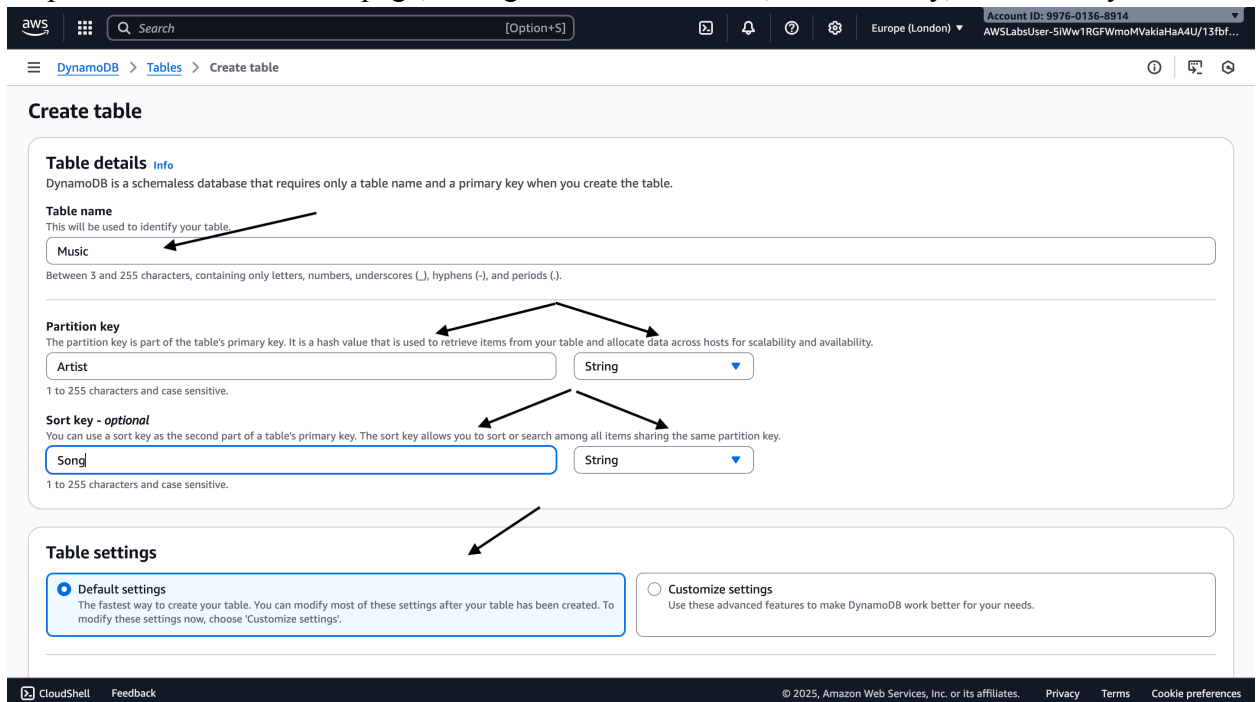


Figure 3: Table details (source: personal collection)

- Step 4: Finally, Create table

The screenshot shows the AWS DynamoDB console interface. At the top, a green banner indicates "The Music table was created successfully." Below this, the "Tables (2)" section is visible, showing a list of tables. The table "Music" is highlighted, and its status is "Active". A green arrow points to the "Active" status. The table details show a partition key of "Artist (S)" and a sort key of "Song (S)".

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode	Write capacity mode
LabEnforcer_Table	Unavailable					Unavailable	☆		
Music	Active	Artist (S)	Song (S)	0	0	Off	☆	On-demand	On-demand

Figure 4: Music table created (source: personal collection)

Task 2: Add Data

- Step 1: Create an item by configuring the “Artist” Value as “Pink Floyd”,” Song” value as “Money”.
 - Then Create additional attribute by configuring the “Attribute name” as Album, “Value” as “The Dark Side of the Moon”.
 - Create a number attribute by configuring the “Attribute name” as “Year”, and “Value” as “1973”

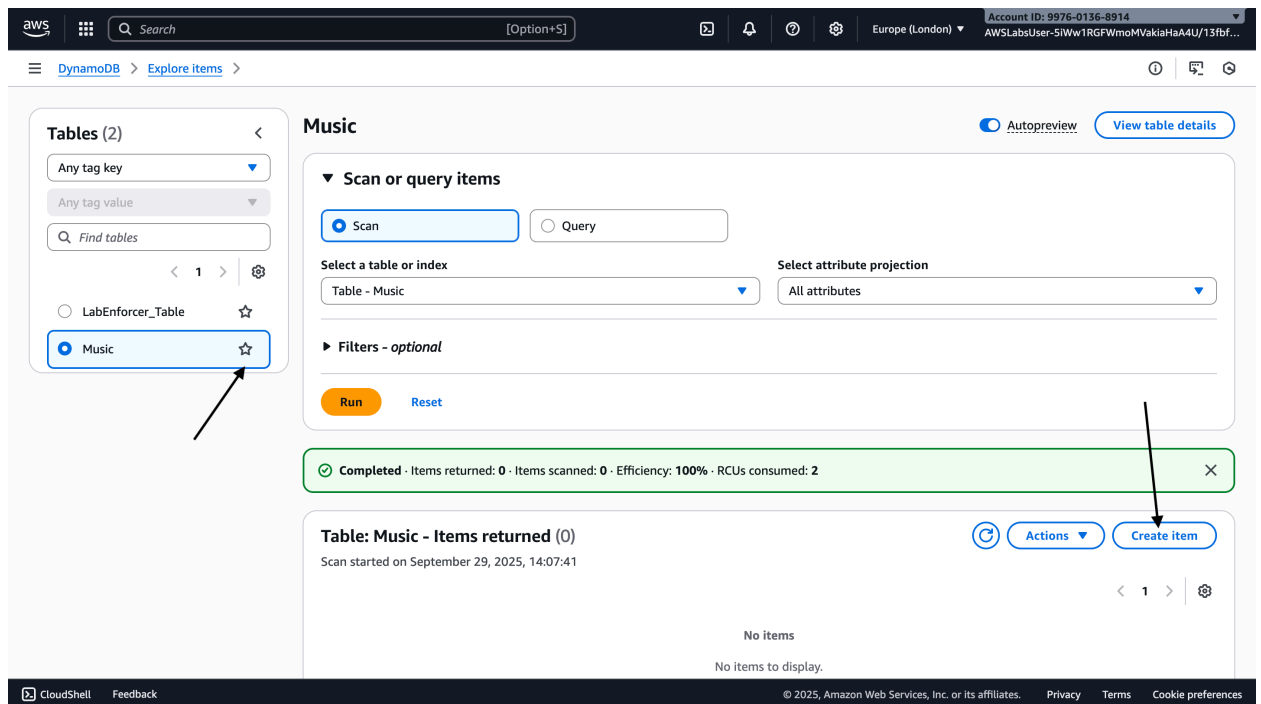


Figure 5: Explore item (source: personal collection)

Create item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Attributes Add new attribute

Attribute name	Value	Type	
Artist - Partition key	Pink Floyd	String	
Song - Sort key	Money	String	
Album	The Dark Side of the Moon	String	Remove
Year	1973	Number	Remove

Cancel Create item

Figure 6: New attribute configuration (source: personal collection)

- Step 2: Create item

Music Autopreview View table details

Scan or query items

☒ Scan ☐ Query

Select a table or index: Table - Music

Select attribute projection: All attributes

Filters - optional

Run Reset

Completed - Items returned: 1 - Items scanned: 1 - Efficiency: 100% - RCUs consumed: 2

Table: Music - Items returned (1) Actions Create item

Scan started on September 29, 2025, 14:16:38

Artist (String)	Song (String)	Album	Year

Figure 7: Attribute created successfully (source: personal collection)

- Step 3: Create a second item with following details:

Attribute Name	Attribute Type	Attribute Value
Artist	String	John Lennon
Song	String	Imagine
Album	String	Imagine
Year	Number	1971
Genre	String	Soft rock

The screenshot shows the AWS Management Console interface for creating a new item in a DynamoDB table. The breadcrumb navigation indicates the path: **DynamoDB** > **Explore items: Music** > **Create item**. The page title is **Create item**, and there are tabs for **Form** and **JSON view**. Below the title, a message states: "You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)".

The **Attributes** section contains a table with the following data:

Attribute name	Value	Type	
Artist - Partition key	John Lennon	String	
Song - Sort key	Imagine	String	
Album	Imagine	String	Remove
Year	1971	Number	Remove
Genre	Soft rock	String	Remove

At the top right of the attributes table is a button **Add new attribute** with a dropdown arrow. At the bottom right of the attributes section are two buttons: **Cancel** and **Create item**.

Figure 8: Additional item (source: personal collection)

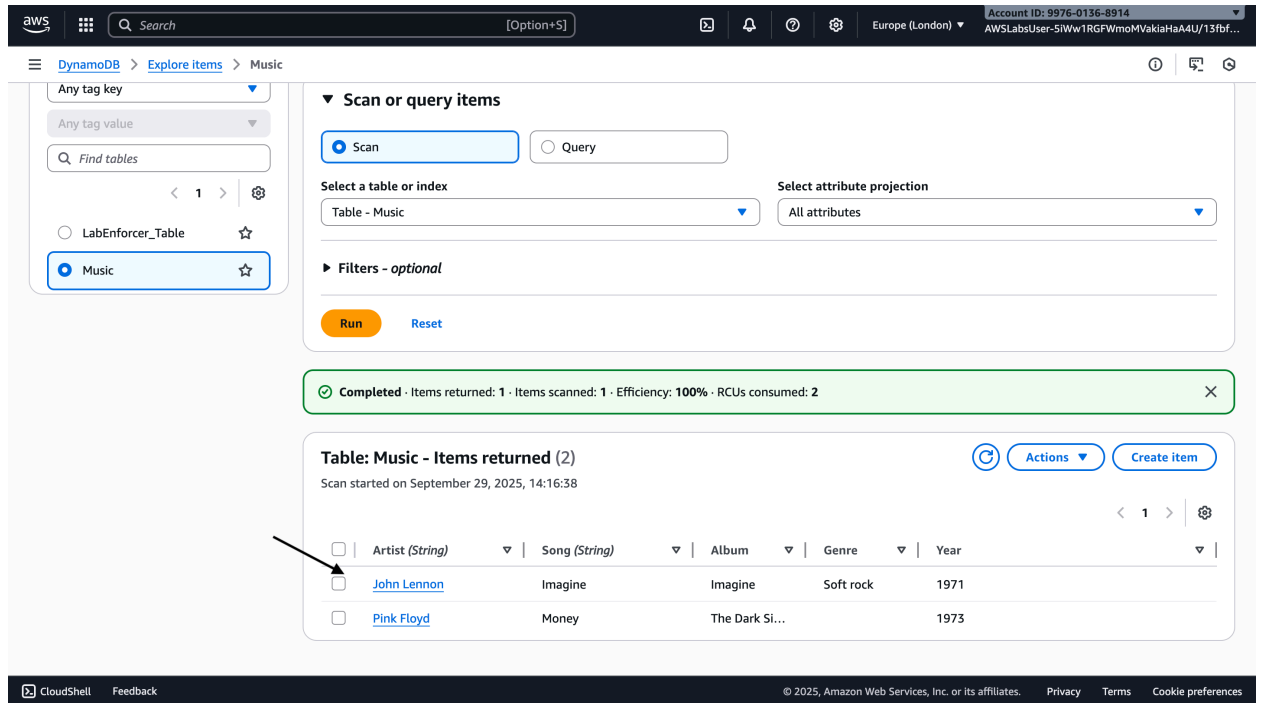


Figure 9: Additional item created (source: personal collection)

- Step 4: Create a third item as follows:

Attribute Name	Attribute Type	Attribute Value
Artist	String	Psy
Song	String	Gangnam style
Album	String	Psy 6 (Six Rules), Part 1
Year	Number	2011
LenghtSeconds	Number	219

Figure 10 shows the AWS Management Console interface for the 'Music' table in a DynamoDB instance. The interface displays the 'Scan or query items' section, where a scan operation was successfully completed. The scan results show one item returned, with the following details:

Artist (String)	Song (String)	Album	Genre	LengthSeconds	Year
Psy	Gangnam Style	Psy 6 (Six R...		219	2011

The scan operation was completed on September 29, 2025, at 14:16:38. The scan results show 1 item returned, 1 item scanned, 100% efficiency, and 2 RCUs consumed. The 'Music' table is selected in the left sidebar, and the 'Scan' button is highlighted in the 'Scan or query items' section. An arrow points to the 'Psy' item in the scan results table.

Figure 10: Psy item created successfully (source: personal collection)

Task 3: Modify an Existing Item

- Step 1: Modify “Psy” by changing the year from 2011 to 2012

The screenshot shows the AWS DynamoDB console interface. On the left, the 'Music' table is selected. The main area shows the 'Scan' results for the 'Music' table. The 'Psy' item is selected, and the 'Edit item' dropdown menu is open. The 'Year' attribute is highlighted in the table.

Artist (String)	Song (String)	Album	Genre	Year
Psy	Gangnam Style	Psy 6 (Six Rules, Part 1)	Soft rock	2012
John Lennon	Imagine	Imagine	Soft rock	1971
Pink Floyd	Money	The Dark Side of the Moon	Soft rock	1973

Figure 11: Editing Psy item (source: personal collection)

The screenshot shows the 'Edit item' page for the 'Psy' item. The 'Attributes' section is visible, showing the 'Year' attribute set to 2012. The 'Save and close' button is highlighted.

Attribute name	Value	Type
Artist - Partition key	Psy	String
Song - Sort key	Gangnam Style	String
Album	Psy 6 (Six Rules), Part 1	String
LengthSeconds	219	Number
Year	2012	Number

Figure 12: Edit item page (source: personal collection)

- Save changes

aws Search [Option+S] Europe (London) Account ID: 9976-0136-8914 AWSLabUser-SiWw1RGFWmoMVakiaHaa4AU/13fbf...

DynamoDB > Explore items > Music

Any tag value

Find tables

LabEnforcer_Table

Music

Scan Query

Select a table or index Table - Music

Select attribute projection All attributes

Filters - optional

Run Reset

Completed - Items returned: 1 - Items scanned: 1 - Efficiency: 100% - RCUs consumed: 2

Table: Music - Items returned (3)

Scan started on September 29, 2025, 14:16:38

Artist (String) Song (String) Album Genre LengthSeconds Year

Psy Gangnam Style Psy 6 (Six R... 219 2012

John Lennon Imagine Imagine Soft rock 1971

Pink Floyd Money The Dark Si... 1973

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Figure 13: Successfully edited the item Psy (source: personal collection)

Task 4: Query the Table

- Step 1: From the navigation pane, choose “Explore items”
 - Select “Music table”
 - Expand “Scan or query items”
 - Select Query
 - For Artist (partition key): use “Psy”
 - For Song (sort key): use “Gangnam Style”

The screenshot shows the AWS DynamoDB console interface for configuring a query. On the left, the navigation pane shows the 'Music' table selected. The main panel is titled 'Scan or query items' and has two tabs: 'Scan' and 'Query', with 'Query' being the active tab. Below the tabs, there are several configuration fields: 'Select a table or index' is set to 'Table - Music'; 'Select attribute projection' is set to 'All attributes'; 'Partition key: Artist' is set to 'Psy'; 'Sort key: Song' is set to 'Gangnam Style' with the condition 'Equal to' and 'Sort descending' unchecked. Below these fields is a section for 'Filters - optional' with a table for adding filters. At the bottom of the configuration section are 'Run' and 'Reset' buttons. A green status bar at the bottom of the configuration area indicates 'Completed · Items returned: 1 · Items scanned: 1 · Efficiency: 100% · RCUs consumed: 2'. The footer of the console shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services.

Figure 14: Query configuration (source: personal collection)

- Then choose “Run”

The screenshot shows the AWS DynamoDB console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and account information (Account ID: 9976-0136-8914, AWS Labs User-SIWw1RGFWmoMVakiaHaa4U/13fbr...). Below the navigation bar, the breadcrumb trail indicates the path: DynamoDB > Explore items > Music.

The main content area shows the query configuration and results. The 'Sort key' is set to 'Song' with a condition of 'Equal to' and a value of 'Gangnam Style'. There are no filters applied. The 'Run' button is highlighted in orange. Below the configuration, a green status bar indicates the query is 'Completed' with 1 item returned, 100% efficiency, and 0.5 RCUs consumed.

The results section, titled 'Table: Music - Items returned (1)', shows a table with the following data:

Artist (String)	Song (String)	Album	LengthSeconds	Year
Psy	Gangnam Style	Psy 6 (Six R...	219	2012

Figure 15: Query run results (source: personal collection)

- Step 2: Select “Scan”
 - For “Attribute name”: enter “Year”
 - For “Type”: select “Number”
 - For “Condition”: select “Equal to”
 - For “Value”: 1971

Figure 16 shows the AWS DynamoDB console interface for setting up a scan operation. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for "DynamoDB" and "Explore items". Under "Explore items", there are filters for "Any tag key" and "Any tag value". A search bar labeled "Find tables" is present. Below this, there are two tables: "LabEnforcer_Table" and "Music". The "Music" table is selected, indicated by a blue highlight and a star icon.
- Scan or query items section:** This section is titled "Scan or query items". It has two radio buttons: "Scan" (selected) and "Query". Below these, there are two dropdown menus: "Select a table or index" (set to "Table - Music") and "Select attribute projection" (set to "All attributes").
- Filters - optional section:** This section allows for adding filters. It has a table with columns: "Attribute name", "Condition", "Type", and "Value". A filter is added for "Year" with the condition "Equal to" and the value "1971". The "Type" is set to "Number". There is an "Add filter" button and a "Remove" button.
- Run button:** A prominent orange "Run" button is located below the filters section. An arrow points to it from the left.
- Table: Music - Items returned (1):** This section shows the results of the scan. It includes a "Query started on" timestamp (September 29, 2025, 15:02:15). Below this is a table with columns: "Artist (String)", "Song (String)", "Album", "LengthSeconds", and "Year". The table contains one item: "Psy" (Artist), "Gangnam Style" (Song), "Psy 6 (Six R..." (Album), "219" (LengthSeconds), and "2012" (Year). There are "Actions" and "Create item" buttons to the right of the table.

The bottom of the interface shows the AWS footer with "CloudShell", "Feedback", and copyright information for Amazon Web Services, Inc. or its affiliates. There are also links for "Privacy", "Terms", and "Cookie preferences".

Figure 16: Scan set up (source: personal collection)

○ Lastly, “Run”

The screenshot shows the AWS DynamoDB console interface. In the left sidebar, the 'Music' table is selected. The main area has the 'Scan' tab active. Below the tabs, 'Table - Music' is selected for the table and 'All attributes' for the attribute projection. A filter is applied for 'Year' equal to '1971'. The 'Run' button is highlighted with an arrow. Below the configuration area, a green status bar indicates the scan is 'Completed' with 1 item returned, 3 items scanned, 33.33% efficiency, and 2 RCUs consumed. Below this, the 'Table: Music - Items returned (1)' section shows a table with columns: Artist (String), Song (String), Album, Genre, and Year. The first row contains the data: John Lennon, Imagine, Imagine, Soft rock, 1971. An arrow points to the 'Artist' column header.

Artist (String)	Song (String)	Album	Genre	Year
John Lennon	Imagine	Imagine	Soft rock	1971

Figure 17: Scan successful (source: personal collection)

Task 5: Delete the Table

- Select “Music table” from the navigation pane
- Choose “Delete”

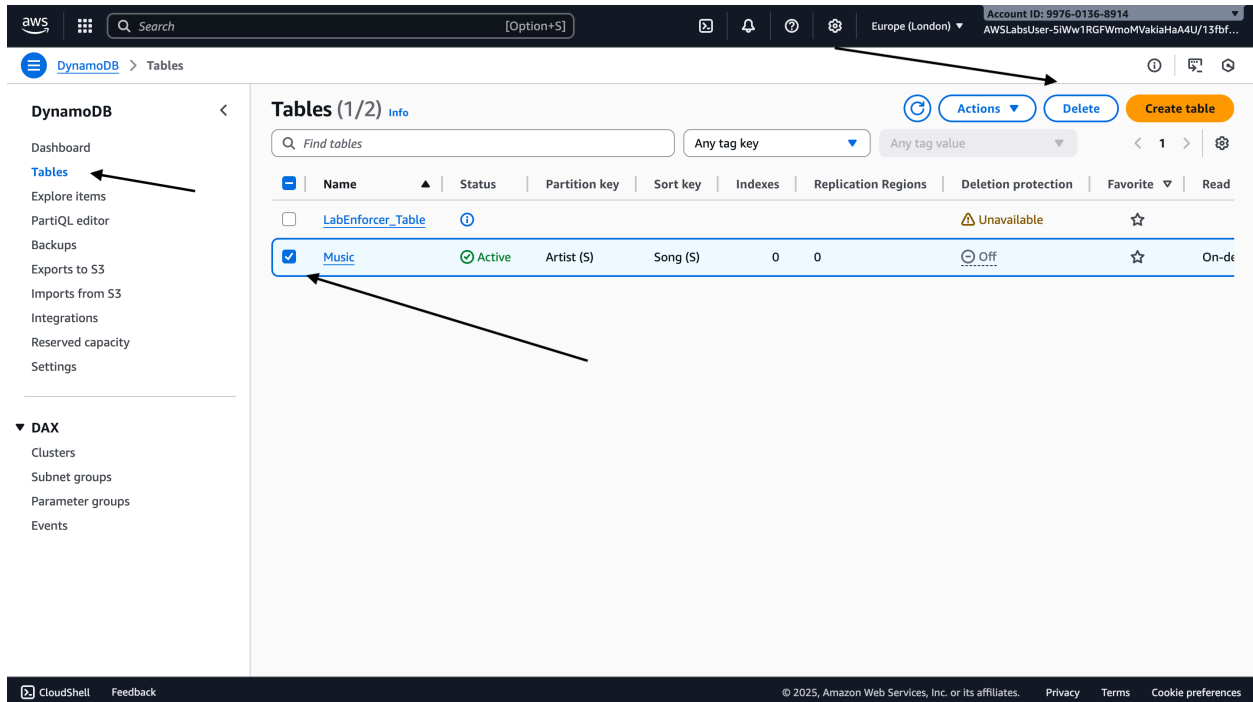


Figure 18: Tables page (source: personal collection)

- From the pop-up window, “Enter confirm to agree”

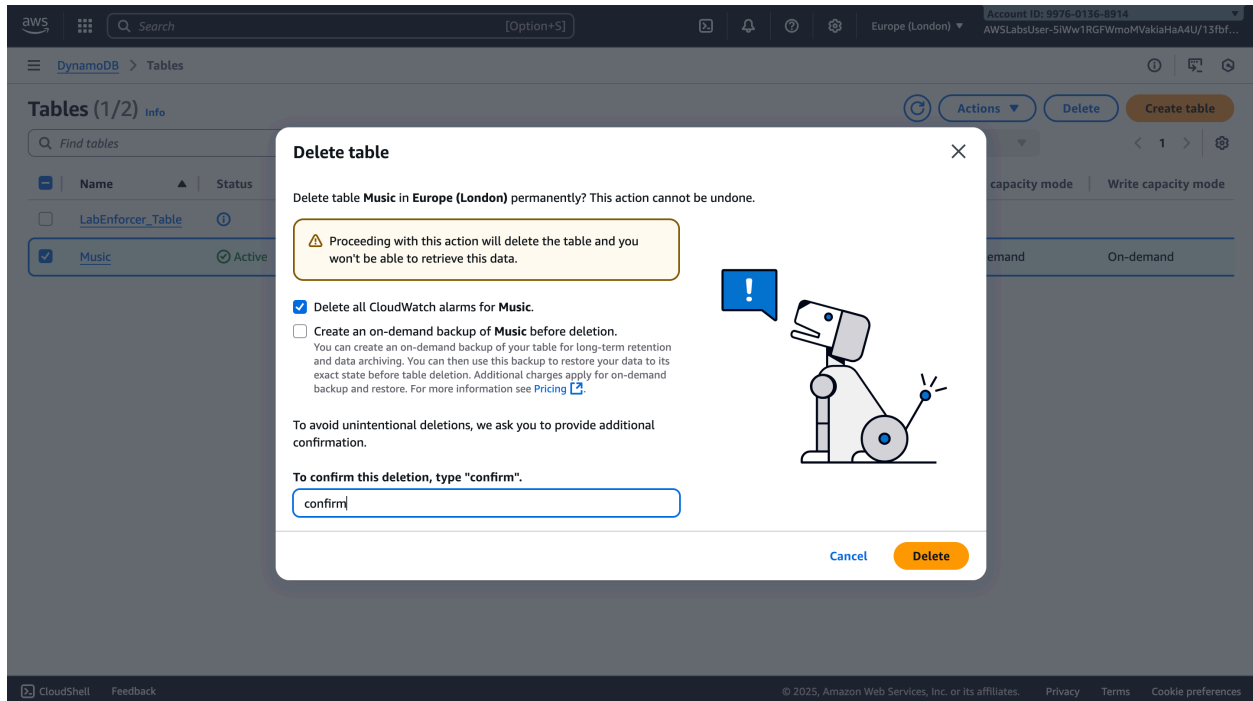


Figure 19: Pop-up message (source: personal collection)

○ Choose “Delete”

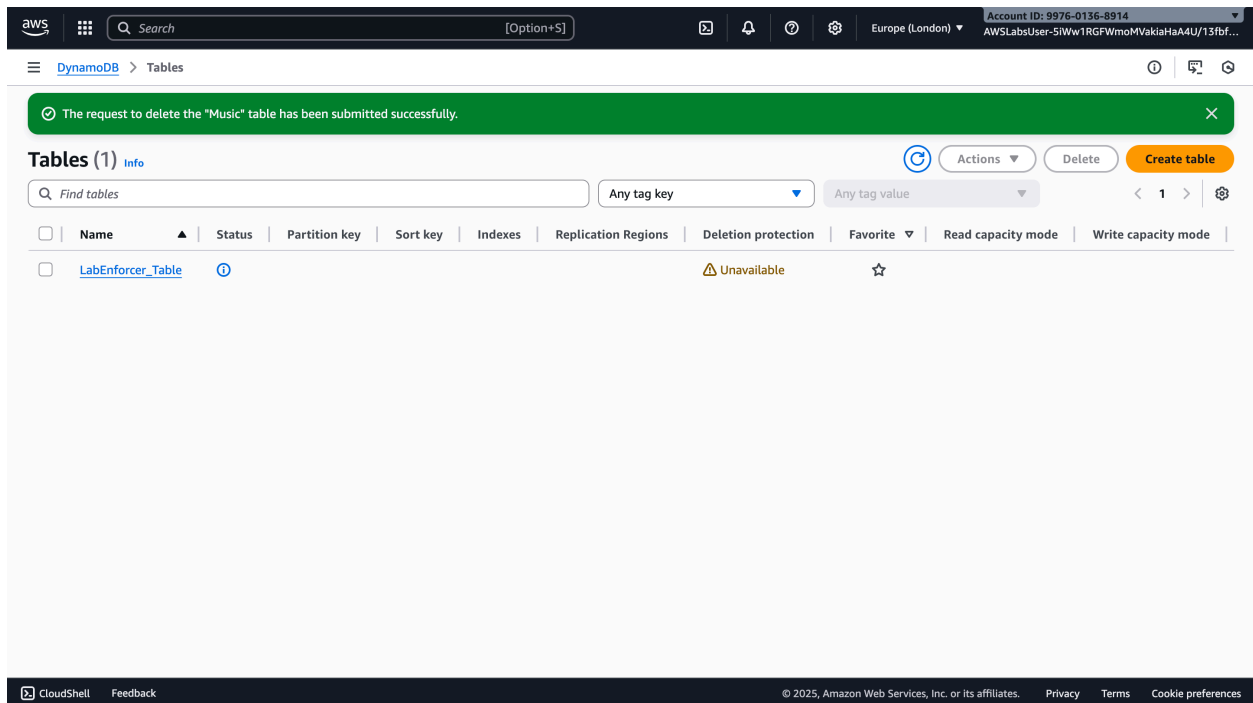


Figure 20: Table Deleted (source: personal collection)

Conclusion

In this lab, I learned to use DynamoDB to create tables, manage data, run queries, and delete tables. I also gained a better understanding of item creation, attribute configuration, and the use of partition and sort keys.

Reference

AWS Skill Builder. Introduction to Amazon DynamoDB (n.d). Retrieved from <https://lab.builder-labs.skillbuilder.aws/sa/lab/arn%3Aaws%3Alearningcontent%3Aus-east-1%3A470679935125%3Ablueprintversion%2Fspl-71%3A3.0.17-55892e9a/en-US>